EXCEL LEARNING SERIES

Introduction to Data Analysis Using Excel

Turn Data into Insights

Learn • Explore • Analyze • Make Better Decisions

Work with Data
Clean, organize and prepare

Explore with Charts
Visualize your data

Analyze with Formulas
Find valuable insights

Build Dashboards
Present your results







Said Fawzy

LEARN • PRACTICE • BUILD

From Data to Insights

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Introduction

An organization’s true competitive edge lies in the agility of its decision-making—responding swiftly to market fluctuations, outpacing competitors, and driving sustainable growth by maximizing return on investment, optimizing costs, and identifying internal strengths, weaknesses, and market dynamics.

Data and information serve as the backbone of every sound organizational decision. However, raw data alone cannot steer leadership in the right direction. It must directly answer precise business questions, delivering clarity through well-crafted visual reports, interactive charts, and intuitive dashboards.

Accordingly, I created this course and handbook as an accessible, focused introduction to data analysis and executive reporting using Microsoft Excel. It is tailored for aspiring analysts and professionals seeking a solid, practical grounding in core reporting workflows, structured data handling, and essential analytical tools provided by Excel.

The curriculum is structured around a progressive, hands-on learning framework:
*** **Before We Start** (Foundations):** Establishing chart visualization fundamentals and core plotting techniques (Lab 0).
*** **Module 01 - Reporting in Excel**:** Transitioning from raw numbers to meaningful executive visual reports, paired with practical demonstrations and knowledge-check reviews.
*** **Module 02 - Excel Tables**:** Harnessing structured tables, automatic range management, and clean data organization (Lab 02).
*** **Module 03 - Basic PivotTables & Interactive Dashboards**:** Unlocking the power of PivotTables to summarize dense records, perform multi-dimensional cross-tabulation, and assemble interactive dashboards (Labs 3A & 3B).

Each module integrates:
1. A conceptual walkthrough accompanied by a live step-by-step demonstration.
2. Knowledge-check review questions to ensure mastery of core principles.
3. Hands-on practical lab exercises using multi-year business scenarios, enabling learners to appreciate the real-world value of every calculation, metric, and visual created.

As this handbook serves as the first milestone in the data analyst's journey, training datasets have been cleanly pre-structured for immediate charting and reporting without friction. Advanced data ingestion pipelines, complex ETL transformations, and relational data modeling have been intentionally reserved for subsequent advanced tracks.

I sincerely hope this material clarifies the philosophy of analytical reporting, demystifies data storytelling, and serves as an inspiring launchpad for everyone embarking on this career path.

May Allah grant us success in all that He loves and is pleased with.

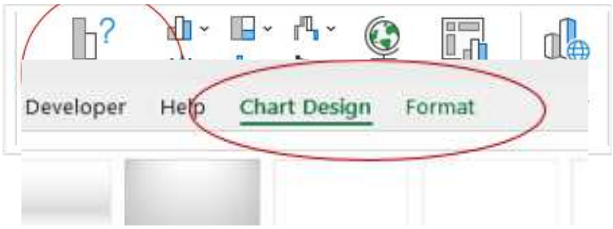
Said Fawzy
Information Center Manager
Tenders Department — The Arab Contractors
Cairo, April 1, 2023

Before We Start: Excel Chart Basics

Let us review the basics of creating charts in Excel.

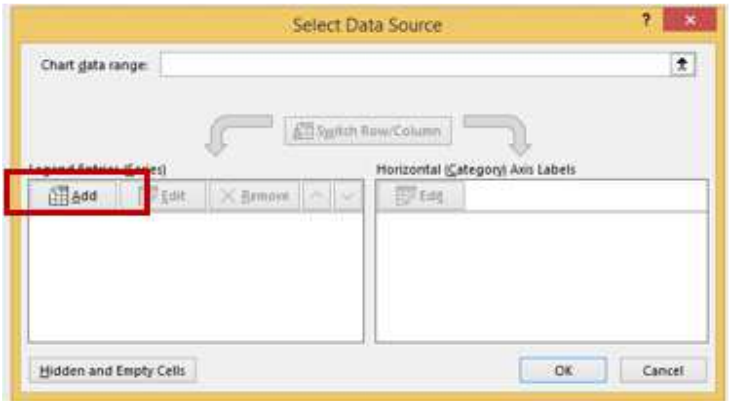
How to Insert a Chart

The easiest way is to use Insert ➔ Chart Group ➔Recommended Chart



You will have two new taps when you insert chart

You can add data series manually using

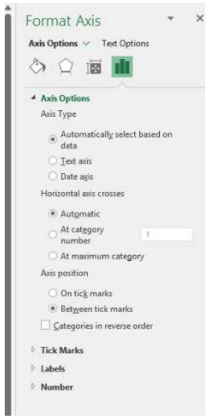


Adjusting the chart

You can add and remove chart element

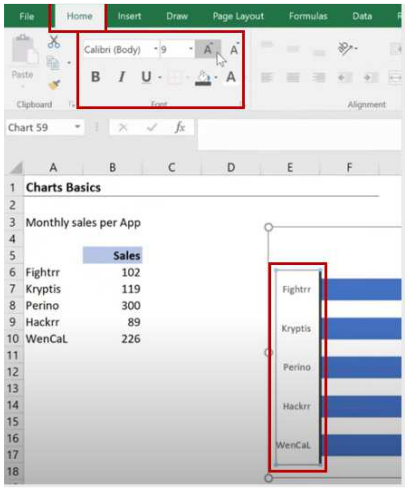


You can format any element



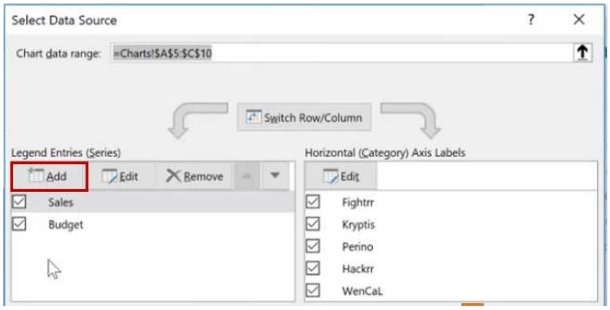
Improving Chart

Change Axis text fonts and color



Adding More Series





Lab 0: Basics of Charts

Use_Lab 00_Chart Basics_Start.xlxs from Labs Folder.

You can use the step by step available in the file

Excel Charts Basic.pptx

Step 1 Inserting a chart :

We want to visualize this table of data

1. 1-Insert a chart using Insert Recommend Charts.
2. Select a cell outside the data table , and Insert a chart Using Bar chart option.
3. Now you have an empty chart.
4. Notice you have new taps.
5. Add data to the new chart from the table.
6. Now you have another chart you made manually
7. Delete all charts you have created.
8. Create a default chart using Alt + F1
9. Change the Default chart to line chart.
10. Delete the current chart
11. Try your new default chart.

Step 2 Adjusting the chart :

1. Delete all charts you have created

2. Insert a cluster bar chart
3. Go and check data elements using the + icon.
4. Add data labels
5. Change where data labels appear.
6. Click More options
7. Right click data series and format the element.
8. Reverse the Y Access.
9. Change the Gab width between bars.
10. Select and delete the Grid line

Step 3 Improving the chart :

1. Enlarge Y Access line and increase its font and make it bold.

Step 4 Adding more series :

1. Add another column to the table Budget.
2. Make budget = 1.1 X Sales.
3. Select the Chart and expand data series.
4. Add the same series again with different name using right click the data series.
5. Change the series order.
6. Remove the new series.

Step 4 Create combination Chart :

Convert Budget to become line chart

1. Convert Budget line into Dash points
2. Change time size color of the dash
3. Add data labels whit in the bottom of columns of sales.
4. Add data labels on top for Budget
5. Add title to the chart
6. Add legend to chart
7. Delete Y axis .

Module 01: Introduction to Reporting In Excel

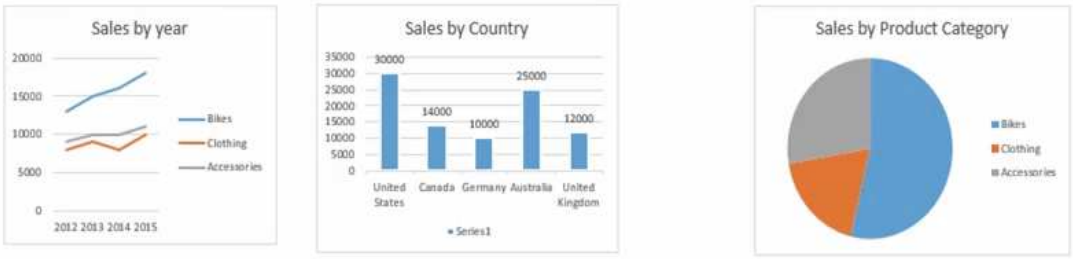
You are now working for Mozilla Bicycle Company. They are selling Bicycles and their components.



Demo 1: Creating Charts

Lucy is a new Sales Manager. She wants to know information about Company sales.

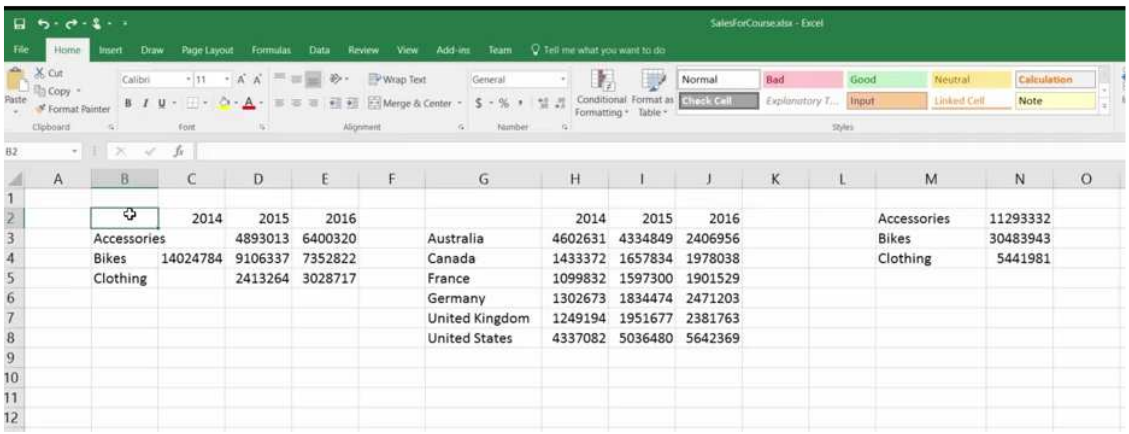
You think you want to show her :



And you think it is enough to show her the performance of the company.

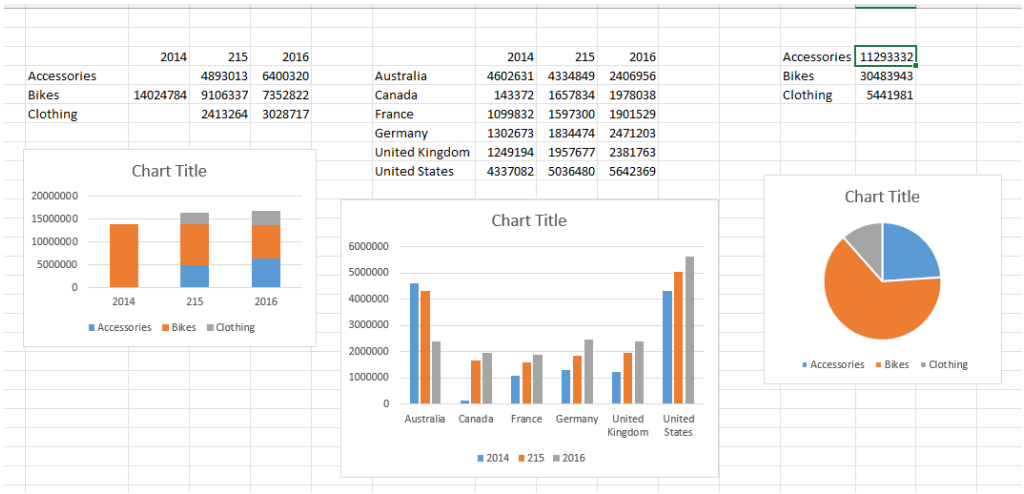
You contacted Jack your friend in IT and he prepared the data for you.

Jack sends you this file



		2014	2015	2016			2014	2015	2016			Accessories	11293332	
	Accessories		4893013	6400320		Australia	4602631	4334849	2406956			Bikes	30483943	
	Bikes	14024784	9106337	7352822		Canada	1433372	1657834	1978038			Clothing	5441981	
	Clothing		2413264	3028717		France	1099832	1597300	1901529					
						Germany	1302673	1834474	2471203					
						United Kingdom	1249194	1951677	2381763					
						United States	4337082	5036480	5642369					

You created this report to Lucy



But she needs more details and many other ways to represent the data.

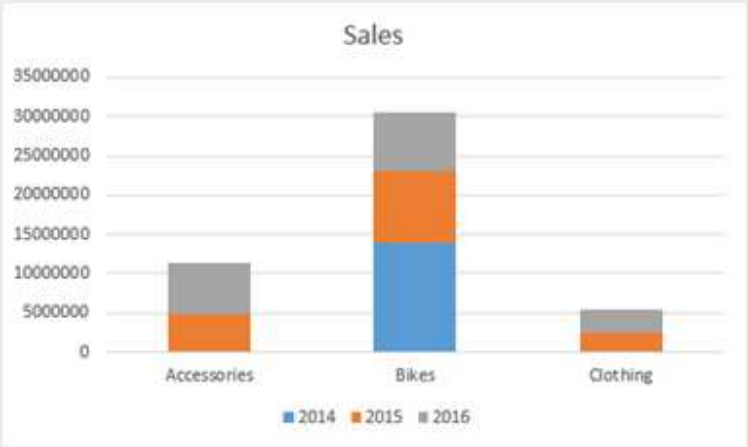
She is interested in:

- profitability,
- subcategories,
- know more about the customers.
- can we classify customers by Age groups? …. etc.

Every time you must go to Jack to give you summery of the data you want. By time Jack Got tired. Jack said I think you need something else; I will prepare for you a set of detailed data and you can create any report or summary you want. All you must do is using **Pivot tables** and **Tables**.

Knowledge Checks

Question 1



You are asked to display a trend for each product by year. What should you do?

- Change the chart type to pie chart.
- Modify the chart to have the product categories as Legend and years as horizontal axis.
- Transpose the table and create the chart from scratch.
- Add summary for each year and insert a new column chart.

Question 2

You have collected the following data and are considering using a pie chart to display it.

	2014	2015	2016
Australia	4602631	4334849	2406956
Canada	1433372	1657834	1978038
France	1099832	1597300	1901529
Germany	1302673	1834474	2471203
United Kingdom	1249194	1951677	2381763
United States	4337082	5036480	5642369

What is a problem that you are likely to encounter by using a pie chart?

- You will have to put the legend below or above the pie because the county names are too long to appear next to the appropriate slice
- The differences in the data are too small between Canada, Germany, and the United Kingdom for the pie chart to have value.
- The pie chart will only show one of the year's data.
- The numbers collected in the data exceed those that can be used effectively in a pie chart.

Question 3

You have collected the following data and want to use a chart to display it.

	2014	2015	2016
Australia	4602631	4334849	2406956
Canada	1433372	1657834	1978038
France	1099832	1597300	1901529
Germany	1302673	1834474	2471203
United Kingdom	1249194	1951677	2381763
United States	4337082	5036480	5642369

Which chart should you use to display it?

- Histogram
- Line chart.
- Pie chart
- Scatter plot

Question 4

You are asked to display the percentage of sales by category for each year in a column chart.

	2014	2015	2016
Accessories		4,893,013	6,400,320
Bikes	14,024,784	9,106,337	7,352,822
Clothing		2,413,264	3,028,717

Which of the following chart types should you use?



- A
- A and B
- B and C
- C

Lab 01:

Scenario:

Use Data in file Lab1Start.xlsx

You are the new marketing manager of an established Bicycle company. The company sells bicycles and accessories, such as clothing and other accessories to bikers in six countries.



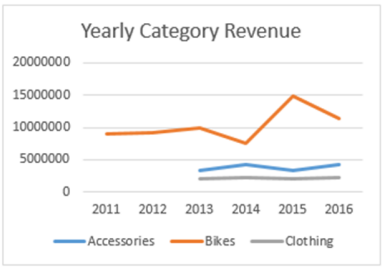
The company has just hired Lucy as its new Sales manager. You are tasked to introduce Lucy to the company, its product portfolio, and its sales performance since 2011. To do this, you have asked Jack, the IT manager to prepare some data for you.

Now, it's your job to present this data in a compelling manner.

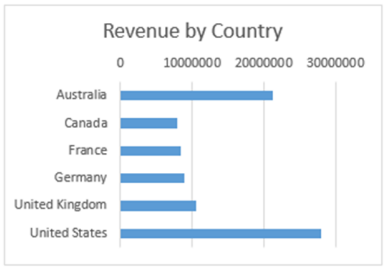
Step 1:

The first thing you'd like to do is to present the data graphically. You do this by creating charts for the three groups of data you received from Jack, your IT manager. If you haven't done so, give the chart a descriptive title.

Your first solution should look like this (numbers may not be the same).







Step 2:

Once you have created the three charts above, let's customize them a little bit more.

For the "Yearly Category Revenue", display the total for each year as a separate line.

1- Create a total line in the range B7 to H7.
2- Select the **Yearly Category Revenue** chart.
3- The data source for the chart is highlighted.
4- Drag the highlighted selection to include the total row (B7 to H7)

Question 1

What does the Yearly total sales look like?

- Increasing year over year.
- Similar to the yearly bike's revenue
- Decreasing year over year.
- Similar to the yearly accessories' revenue

Step 3:

For the "Revenue by Category", display the percentage for each component.

1. Select the **Revenue by Category** chart.
2. Select the pie in the chart, right-click and add data labels.
3. Select **Percentage** in the **Label Options**.

Question 2

What percentage of the total revenue comes from the **Bikes** category?

- 10%.
- 18%
- 36%.
- 72%.

Step 4:

For the **Revenue by Country**, sort the countries from the one with highest revenue to lowest revenue.

1. Select the data source for the **Revenue by Country** chart.
2. Sort the data by **Revenue** from **Largest to Smallest**.
3. Select the **Revenue by Country** chart.
4. Select the Y axis, the axis that shows the country name.
5. Select **Categories in reverse order** in the **Axis options**.

Question 3

Rank the Countries from the highest to lowest revenue. (Highest in Zone 1, Lowest in Zone 6)



Module 02: Excel Tables

Jack prepared us an excel file with the data needed. He said try to use the Excel tables. So what is it ?



Before we start:

Functions you should know:

- IF
- Nested IF
- IFS
- RIGHT
- TEXT
  See Extra Demo Folder

Demo 02 : Excel Tables



Knowledge Checks:

From Quiz folder in Folder Mod 02 use this file :
SalesForCourse_quizz_table.xlsx.

Question 1

What is the difference between a table and a range in Excel? Select four that apply.

- In tables, you can add many types of totals for each column without writing any formulas..
- A range can be converted to a table, but a table cannot be converted to a range.
- Tables are formatted with alternate colors by default.
- Formulas in tables refer to other columns by name and not by regular Excel reference.
- Formulas in table columns are automatically applied to new rows.

Question 2

If you filter the table to show only sales in United Kingdom, what will be calculated in the Total row?

- The total of all rows.
- The totals will recalculate according to the filtered rows.
- The totals will recalculate according to the unfiltered rows.
- The totals will recalculate only when you reopen the file.

Question 3

You are only interested in the data for male customers in France. After setting the appropriate filters, answer the following questions.

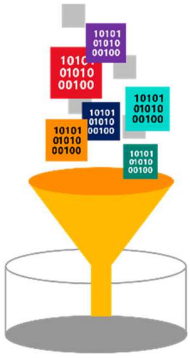
a. How many rows are displayed?
b. What is the total revenue for the displayed rows? Only provide whole number. Do not provide symbols or any decimals places.
c. What is the average revenue? Only provide numbers, no currency symbols, and report to two decimal points.

Lab 02: Creating Excel Table

Use Data in file Lab2Start.xlsx

Scenario

While your first attempt to show the company's performance to Lucy was not bad, clearly, she has a lot more requirements than what you provided. She wants to know more about the year over year sales, sliced into different categories, sub-categories, and countries. She also wants to see additional information such as customer demographics.



Jack has provided you with a different data source. This time the data has more than one hundred thousand rows.

Before you can create additional reports to Lucy, first you need to prepare the data.

Step 1:

The first thing you need to do is to convert the data into an Excel table.

Once you do that, you can add total row, filter the data, and the total will reflect the total only for the filtered data. Let's try this. Add a total row for the table and use the Sum aggregation to show the total of the **Revenue** column and then filter the data only for **United States**.

Question 1

What is the total revenue for all the sales in the **United States**?

- 101990988
- 27975547
- 24606895
- 12494351

Step 2:

Now, you need to add several columns, derived from existing columns in the data.

First, let's add a "Month" column. Insert a new column to the left of the **Customer ID** column, and use formula to derive the month of sales from the **Date** column. HINT: Use the Text() function (=TEXT([@Date],"MM")

Question 2

What is the total revenue for all the sales in the month of **December**?

- 9086931
- 8801313
- 6809175
- 6494188

Step 3:

Next, let's add an "Age Group" column. Insert a new column to the left of the **Customer Gender**, and use formula to derive the age group from the **Customer Age** column. Let's group the customers based on the following criteria:

- Youth (<25)
- Young Adults (25-34)
- Adults (35-64)
- Seniors (>64)

HINT: Use the nested IF() functions. Alternatively, you can use the IFS() function if it is available in your version (2016 + updates from O365)

Question 3

What is the total revenue for all the sales for **Young Adults** Age Group?

- 56053751
- 29918498
- 30655614
- 5556163

Step 4:

Now, let's add a "Frame Size" column. Insert a new column to the left of the **Order Quantity**, and use formula to derive the frame size of a bicycle from the last two characters of the **Product** column, when the **Product Category** is **Bikes**. Otherwise, leave it blank.
HINT: Use the IF() and RIGHT() function.

Question 4

What is the total revenue for all the bikes with frame size **62** for the customer age group **Seniors**?

- 22914
- 125893
- 159102
- 12452

Step 5:

Last but not least, let's add a "Profit" column. Insert a new column to the right of the **Revenue**, and use formula to derive the Profit from both the **Revenue** and **Cost** columns. Show the total for the **Profit** column. Use the Sum aggregation in the total row of the table, for the **Profit** column.
HINT: **Profit** is **Revenue** less **Cost**.

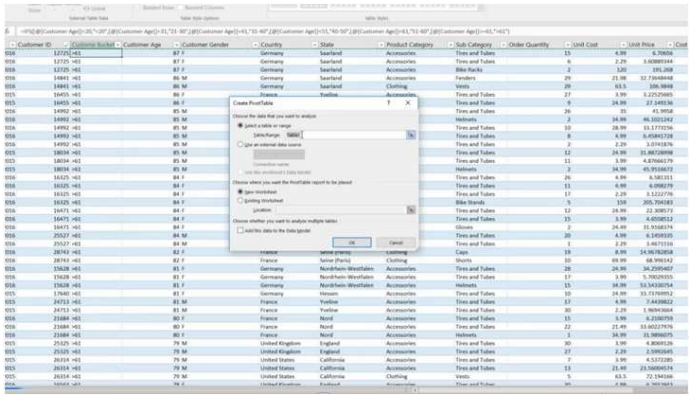
Question 5

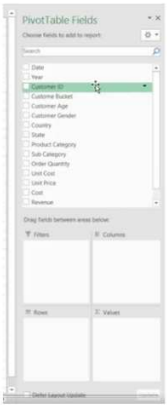
What is the total profit for **United States** sales in the month of **October 2015**, for customer age group **Adults**?

- 111823
- 142036
- 148903
- 138419

Module 03: Basic Pivot Tables

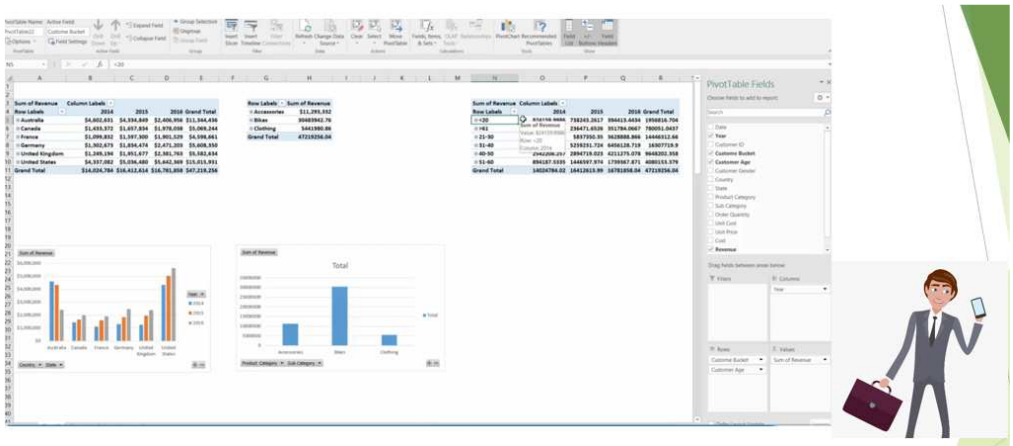
Demo 3A: Basics of Pivot Tables







Demo 3B: Creating Dashboard using Pivot table

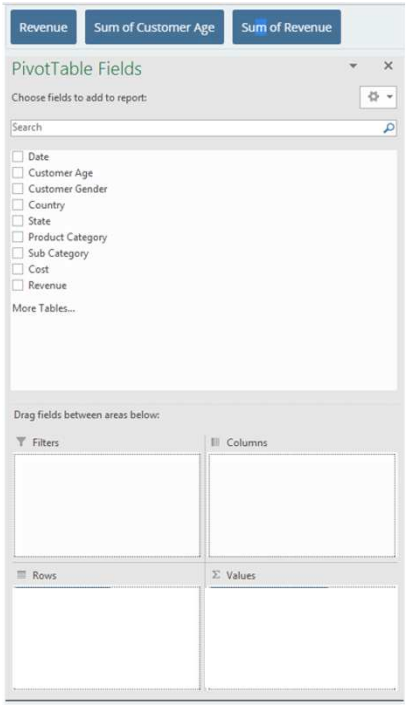


Knowledge Checks

Question 1

PROBLEM

- You have the following Pivot Table fields.
- You need to calculate the number of sales by age.
- Drag the appropriate field names and aggregated field names to the appropriate Pivot Table Areas (Filters, Rows, Columns, or Values) to perform the task



Question 2

PROBLEM

- You have the following Pivot Table fields.
- Your manager asks you to build a pivot table that shows revenue by state and country.
- How would you arrange the field names and aggregated field names to the appropriate Pivot Table Areas (Filters, Rows, Columns, or Values)?



Question 3

You have a pivot that is based on an Excel table. You need to add new rows to the Excel table and want to be able to see the values for these new rows in the PivotTable. What should you do?

- Create a new pivot table.
- Add rows anywhere in the table, and it will automatically update the pivot to include the new values.
- Add rows anywhere in the table, and refresh the pivot table.
- You must add the new rows only after the last row and refresh the pivot table.

Use Data in file SalesForCourse_quizz_table.xlsx to answer

Question 4

Using the data provided, create a pivot with “Country “as rows, “Month” as columns, and “Revenue” as values. Drag “Years” to the filter area. What does the value 286,779 in the first summarized field of the pivot table (uppermost left value) represent?

- The total for January in France for the first year (2015).
- he total for January in France for the last year (2016).
- The total for January in all countries.
- The aggregated value for January in France in 2015 and 2016

Question 5…Numeric input

Create a new PivotTable with “Customer Gender” as rows, “Product Category” as columns, and “Revenue” as values.

a. How much revenue was raised through the purchase of bikes by women? Provide a whole number without currency symbols or decimal places.

b. How much revenue was raised through clothing purchased by men? Provide a whole number without currency symbols or decimal places.

Question 6

When you summarize data using an Excel Pivot Table, each number found in a Pivot Table cell represents one or more records in the pivot table's source data and is directly related to the Column and Row fields you used in the Pivot Table. A quick way to review this underlying set of records is to use the Show Details feature - right click within the desired cell and select Show Details from the menu options. This is commonly called **Drilling down into the data** and can also be achieved by double-clicking in the cell in question.

Create a new PivotTable with "Country" as rows, "Product Category" as columns, "Year" as a filter, and "Revenue" as values. Filter by year 2015.

Drill down into the data for the cell that represents the revenue for bike sales in Germany, 2015. A new worksheet is created with a table in it. What is the content of this table?

- The table shows columns for Year, Country, Revenue, and Products
- The table contains all rows in SalesTable corresponding to 2015, Bikes and Germany
- The table includes data from SalesTable for the first month of 2015 and the first subcategory of Bikes.
- The table includes data from SalesTable for 2015 and the first subcategory of Bikes.

Question 7…Numeric input

Calculate the answers to the following questions using the data provided and PivotTables. All answers should be provided in whole numbers, without currency symbols or decimals

a. Total revenue for sales of bikes in the country that sold the most in 2016:

b. Number of sales transactions for bikes in the US in 2016:

c. Average number of clothing items per transaction purchased in the US during 2016 rounded to the nearest whole number:

d. The largest revenue registered in a single transaction in the Germany during 2015:

e. The largest revenue registered in a single transaction during 2015:

Lab 3: Creating Pivot tables

Use Data in file Lab3AStart.xlsx

Scenario:

Now that you have prepared the data in an Excel table, you can start to create pivot tables to aggregate the data and create some reports.

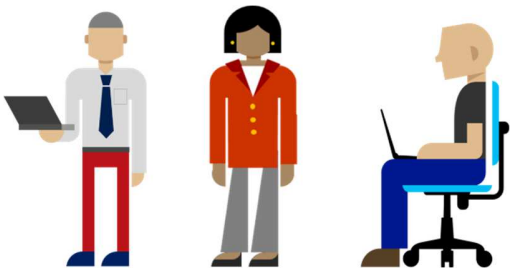
From your conversation with Lucy, you know that she is interested in looking into the yearly sales data broken down by countries, product categories, and age groups.



Lab 3A:

Step 1

First, let's start by naming the Excel table. Name the Excel table prepared in the previous lab to **SalesTable**. From now on, every time you add a pivot table, it should be based on this SalesTable, unless mentioned otherwise.



Now, proceed to add your first pivot table. Insert a new pivot table based on the SalesTable to a new sheet. Arrange the layout so that the pivot table displays the **Product Category** and **Sub Category** in the **Rows**, **Year** in the **Columns**, and **Revenue** (Sum of) as the **Values**.

Question 1

Which year did the company start selling **Touring Bikes**?

- 2011.
- 2012
- 2013.
- 2014

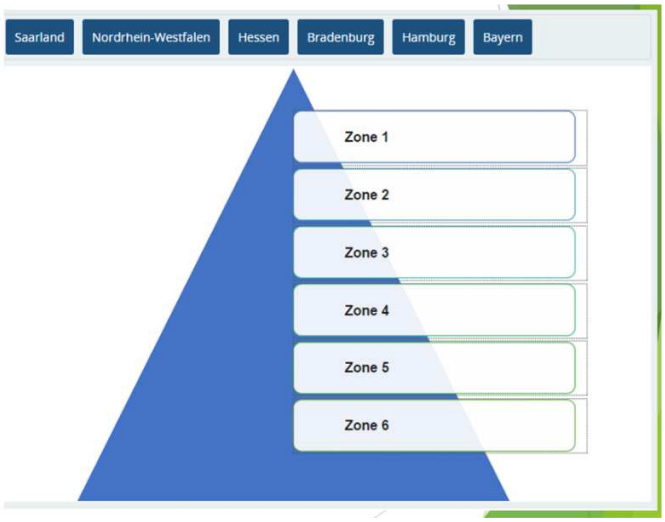
Step 2:

Insert another pivot table to the same sheet, next to the existing pivot table. Arrange the layout so that the pivot table displays the **Country** and **State** in the **Rows**, **Year** in the **Columns**, and **Revenue** (Sum of) as the **Values**. Sort the pivot table by **Sum of Revenue** so that the **Country** and **State** with the highest revenue is displayed first.

Question 2

PROBLEM

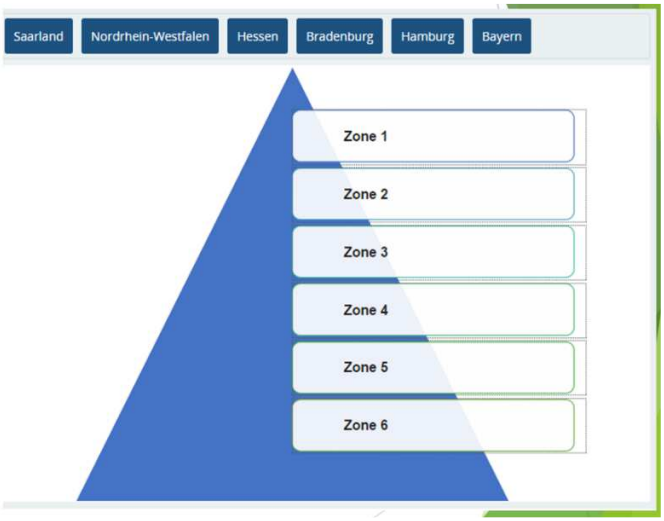
Rank the **States** for **Germany**, from the highest to lowest revenue.
Hint: Sort the **States** field in descending order by **Sum of Revenue**



Question 3

PROBLEM

What about for the year 2013?
Hint: Expand **Germany** and sort the **State** field in descending order by **Sum of Revenue** using values from the 2013 column.



Step 3:

Let's add another pivot table. This time arrange the layout so that the pivot table displays the **Frame Size** in the **Rows** and **Revenue** (Sum of) as the **Values**. Hide the rows that do not have a Frame size (blank Frame size), then sort the

pivot table by **Sum of Revenue** so that the **Frame size** with the highest revenue is displayed first.

Question 4

Which frame size sold the most?

- 40.
- 42
- 44
- 48

Step 4:

Last but not least, add another pivot table with **Age Group** as the **Rows** and **Revenue** (Sum of) as the **Values**. You will learn how to custom sort the Age Group in the next module. But for now, sort the pivot table by **Sum of Revenue** so that the **Age Group** with the highest revenue is displayed first.

Question 5

Which **Age Group** has the lowest revenue?

- Youth.
- Young Adults
- Adults
- Seniors

Lab 3B:

Step 1:

Now you can start adding some charts to the sheet.

First, add a pivot chart for the pivot table that shows yearly sales by Country. Select a **Line** chart to display the yearly trend. Make sure that the **Years** are located in the X axis, the **Revenue** in the Y axis, and the **Countries** as categories. Hint: You might find the Switch Row/Column in Select Data Source window useful. In addition, try Collapsing the Country fields in order to hide the States fields.

Question 1

In this chart, you can clearly see the sales trend for each Country.

Which country's trend is the most different when compared to the other countries?

- United States.
- Australia
- Germany.
- United Kingdom.

Step 2:

Add another pivot chart for the pivot table that shows yearly sales by Product Category. Select a **Column** chart to display the yearly sales by category so that the years are together.

Question 2

Which year does the Bikes category have the least sales?

- 2011.
- 2012
- 2013.
- 2014

Step 3:

Add another pivot chart, this time for the pivot table that shows Revenue by Age Group. Select a **Pie** chart to display the proportion of each **Age Group**. Add the **Data labels** to show percentage, formatted to two decimal points.

Question 3

In this chart, you can clearly see the proportion of sales for each **Age Group**.

What is the proportion of sales for **Young Adults**?

- 54.96%.
- 29.33%
- 35.95%
- 49.74%

Step 4:

Add another pivot chart, this time for the pivot table that shows Revenue by Frame size. Select a **Bar** chart to display the order of sales by **Frame size**. Sort the Y axis to show the **Frame size** that has the most sales on the top.
Hint: Select the **Categories in reverse order** option in the Axis Options.

Question 4

In this chart, you can clearly see the order of sales by Frame size.

Which frame size sold the least?

- 52.
- 54
- 46
- 48

Module 04: Dashboard

Demo 4A: Slicers



Demo 4B: Slicers and Filters



Knowledge Checks:

Use Data in file SalesForCourse_quizz_table.xlsx.

Question 1

Using the data provided, create a pivot with “Country” as rows, “Product Category” as columns, and “Revenue” as values. Slice months by “April.” What does the grand total for revenue (2,200,490) represent?

- Total revenue for April 2015.
- Total revenue for April 2016.
- Total revenue for either 2015 or 2016 depending on which series of data appears first in the data.
- Total combined revenue for April in 2015 and 2016.

Question 2

Which objects can be sliced by slicers? Choose 3 that apply.

- Ranges.
- Pivot Charts.
- Pivot Tables.
- Tables.

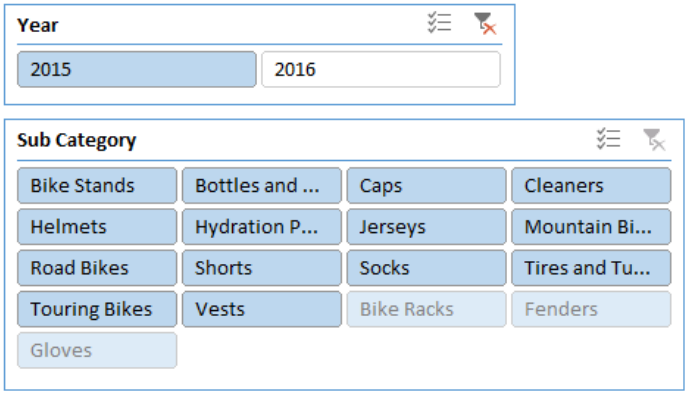
Question 3

When would you use both a slicer and a filter based on the same field?

- Never. Using slicers and filters at the same time might contradict therefore it is not recommended..
- When you want to search items in a field that contains many distinct values.
- When you want to apply filters on multiple Pivot Tables.
- When you want to search items and apply the results on multiple PivotTables

Question 4

You have the following slicers.

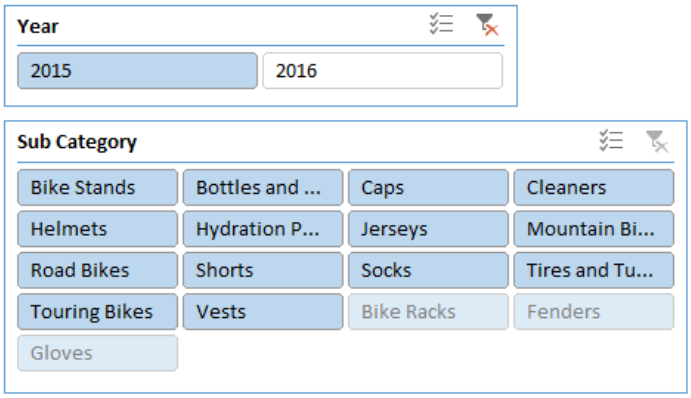


Why are Bike Racks, Fenders and Gloves at the bottom of the Sub Category slicer in a lighter color?

- They contain less rows then the others across both years.
- During 2015, there were fewer sales in these categories than in others.
- During 2015, there were no sales for these sub categories.
- During 2015, the total revenue for these sub categories was negative.

Question 5

You have the following slicers.



What will happen if you click "Bike Racks"?

- All Pivot Tables connected to sub category slicer will be blank.
- All Pivot Tables connected to both year and sub category slicers will be blank.
- This sub category cannot be selected.
- 2016 will become selected in the year slicer above.

Lab 4: Dashboard

Use Data in file Lab4AStart .xlsx.

Scenario:

You have created several pivot tables and pivot charts for Lucy.

So far everything has been well received by her. However, she would like to have easier ways to slice and dice the reports and charts herself.

You sat down with Lucy, and come up with several different ways that Lucy could slice the data

- Year
- Country
- Customer Gender
- Age Group
- Product Category
- Sub Category
- Frame size



Lab 4A:

Step 1:

Start by adding a new sheet named **Dashboard**. Then move (Cut and Paste) the four charts that you have to that sheet. Arrange the charts as appropriate.

Hint: The **Snap to Grid** option located at the **Align Object** on the **Page Layout** tab might be useful.

For easy reference, let's give the charts titles if they don't have any, or rename them as appropriate. Name the four charts as follows:

- Yearly Sales by Country
- Yearly Sales by Category
- Sales by Frame Size
- Sales by Age Group

Hint: To add a chart title, select the chart, then on the **Design** tab, select **Add Chart Element**.

Step 2:

You can now add slicers to the sheet. Select the Yearly Sales by Country chart, and add seven slicers corresponding to Year, Country, Customer Gender, Age Group, Product Category, Sub Category and Frame Size. Arrange the slicers as appropriate.

Step 3 :

The next thing you need to do is to connect these slicers to the charts. Let's do this chart by chart.

Hint: The Filter Connections option is available on the chart Analyze tab.

1. Connect the **Yearly Sales by Country** chart to all slicers, except the **Year** slicer. In another words, disconnect the **Year** slicer from the **Yearly Sales by Country**

2. Connect the **Yearly Sales by Category** chart to the **Age Group**, **Country**, **Customer Gender**, and **Frame Size** slicers.
3. Connect the **Sales by Frame Size** chart to all slicers, except the **Frame Size** slicer.
4. Connect the **Sales by Age Group** chart to all slicers, except the **Age Group**.

For niceties, you can clear the **Gridlines** on the Dashboard,

Hint: The **Gridlines** option is available on the **View** tab.

Once you've done the above, you're ready to present the dashboard to Lucy.

Question 1

A quick glance on the **Yearly Sales by Country** shows that **Australia** has an unusual trend compared to the other countries. Which year does **Australia** have the least sales?

- 2011
- 2012
- 2013.
- 2014

Step 4:

Create an additional pivot chart to show **Sales by Country** using **Pie** chart. Show percentages for each slice of the pie and connect the chart to all the slicers, except the **Country** slicer.

Overall, Australia commands 25% of the company's total sales. But in some of the years, this proportion changes.

Question 2

What is the percentage of **Australia** sales (of total sales) in the year that it has the least sales (previous question)?

- 13%.
- 16%.

- 20%.
- 28%.

Question 3

Let's filter the charts by **Australia** using the **Country** slicer. What might be the cause of this trend?

- The introduction of new line of products in Accessories and Clothing.
- The sharp fluctuations of bicycle sales..
- The sharp fluctuations of Accessories sales..
- The sharp fluctuations of Clothing sales.

Step 5:

Based on your hypothesis (previous question), create an additional pivot chart to show **Sales by Category** using **Pie** chart. Show percentages for each slice of the pie and connect the chart to all the slicers, except the **Category** slicer. Filter the charts by **Australia** using the country slicer and play around with the **Year** filter. Notice for different years, the changes in composition of **Australia** sales by **Category**.

Question 4

For all the years that has all the three categories, which year does the **Bikes** has the least proportion?

- 2013.
- 2014.
- 2015.
- 2016.

Question 5

What is the percentage of **Bikes** sales (of total sales) in that year?

- 48%.
- 59%.
- 66%.
- 75%.

Lab 4B:

Step 1:

Create an additional pivot chart to show **Sales by Customer Gender** using **Pie** chart. Show percentages for each slice of the pie and connect the chart to all the slicers, except the **Customer Gender** slicer.

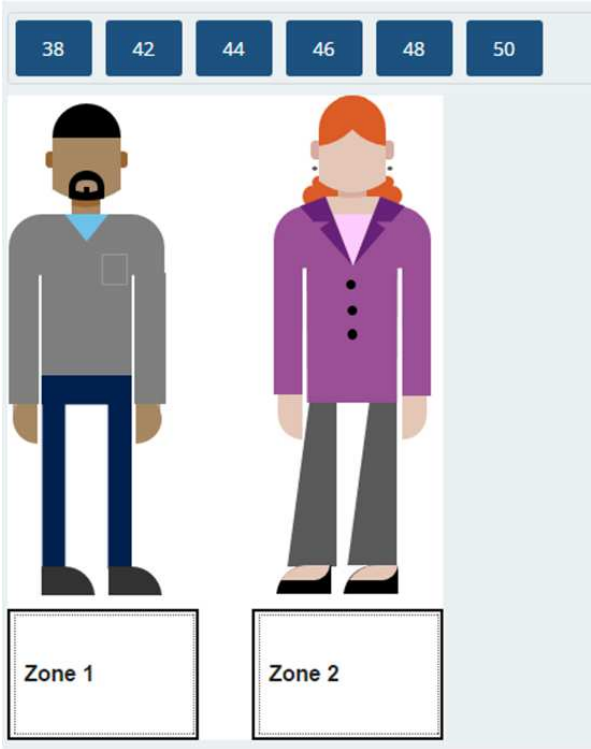
Overall, it is split evenly with 51%:49% of Male to Female ratio.

Question 1

PROBLEM

Which Bike's frame size is more popular for each **Gender**? Place the **Frame size** into the appropriate box.

Hint: Clear any existing filter, then Use the **Customer Gender** filter and review the **Sales by Frame Size** chart.



Step 2:

What about Customer Gender vs age group? Right now the **Sales by Age Group** chart does not differentiate by **Gender**. Modify this chart to be a **Column** chart. Show the **Customer Gender** side-by-side for each age group. Ensure that the chart is connected to all slicers. Last but not least, sort the **Age Group** appropriately.

Hint: Clear any existing filter, place **Customer Gender** as **Legend** and create **Custom List** to sort the **Age Group**.

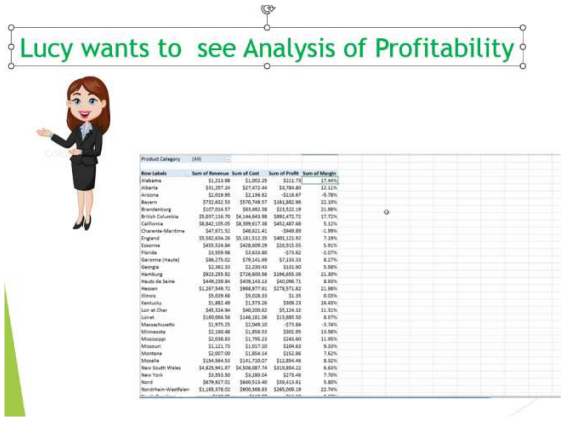
Question 2

In **Australia**, which **Age Group** has more sales revenue to females than to males? (Select two that apply)

- Youth (<25).
- Young Adults (25-34).
- Adults (35-64).
- Seniors (64+).

Module 05: Profitability Analysis and Finding Anomalies:

Demo 5: Profitability analysis:



Conditional Formatting:



Knowledge Checks:

Question 1

You have a sales table that contains “Quantity,” “Unit Price,” and “Unit Cost.” How would you add the following calculations to be used in a PivotTable?

Total Cost

- Add a Pivot Table calculated field (Total Cost = Unit Cost * Quantity).
- Add a calculated column in the table (Total Cost = Unit Cost * Quantity).

Total Revenue

- Add a Pivot Table calculated field (Total Revenue = Unit Price * Quantity).
- Add a calculated column in the table (Total Revenue = Unit Price * Quantity).

Total Profit

- Add a Pivot Table calculated field (Total Profit = Revenue - Cost).
- Add a calculated column in the table (Total Profit = Revenue - Cost).

Profit Margin

- Add a PivotTable calculated field: Profit Margin = (Revenue - Cost) / Revenue.
- Add a calculated column in the table: Profit Margin = (Revenue - Cost) / Revenue.

Question 2

Conditional formatting (Green – Yellow – Red Color Scale) was applied on the above PivotTable. What can you do in order to improve usability of the formatting?

Row Labels	Sum of Revenue	Sum of Margin
Bike Racks	140,854	25%
Bike Stands	150,911	17%
Bottles and Cages	709,407	18%
Caps	255,992	17%
Cleaners	182,635	63%
Fenders	329,204	22%
Gloves	228,353	20%
Helmets	2,738,210	19%
Hydration Packs	403,276	18%
Jerseys	1,834,110	16%
Mountain Bikes	5,176,456	3%
Road Bikes	3,921,989	3%
Shorts	689,184	13%
Socks	61,237	-25%
Tires and Tubes	2,865,915	18%
Touring Bikes	2,387,910	4%
Vests	368,681	16%
Grand Total	**22,444,324**	**10%**

- Replace Green – Yellow – Red Color Scale with Blue – White – Red Color Scale.
- Use Icon Set instead of Color Scale.
- Modify the percentiles for maximum and minimum values.
- Use Data Bars instead of Color Scale.

Lab 5: Profitability Analysis

Scenario

Use Data in file Lab5Start.xlsx.

Lucy was impressed with the dashboard you created.

With the dashboard, she is able to narrow down her interest. Specifically, she is interested with the sales in Australia. She would like to perform simple profitability analysis on the product category and sub-category, for specific year. Furthermore, she wants to have the option of going deeper to product level.



Step 1:

Start by adding a new sheet named **Details**. Add a pivot table that shows **Order Quantity**, **Revenue**, and **Profit** by **Product Category**. Add the **Year** and **Country** field to the filter list.

Add the **Sub Category** field to the rows of the pivot table. Make sure the rows are fully expanded. Now you have quite a few numerical data shown in the pivot table, which might not be as easy to analyze as before. Apply **Data bars** conditional formatting to the **Order Quantity**, **Revenue** and **Profit** column on **Sub Category** level. Use blue, green, and yellow respectively.

Filter the pivot table for **Australia** and **2016**, and answer the following questions.

Question 1

Which **Sub Category** sold the most quantity?

- Tires and Tubes
- Road Bikes
- Helmets
- Bottles and Cages

Question 2

Which Sub Category has the most revenue?

- Tires and Tubes
- Road Bikes
- Helmets
- Bottles and Cages

Step 2:

Now, add a calculated field to your pivot table, name the field **Margin** with the value derived from the **Profit** and **Revenue** column. Format the field as percentage with two decimal places.

HINT: Margin = Profit / Revenue

Question 3

What is the total margin for Australia in the year 2016?

- 38.96%
- 26.26%
- 33.74%
- 20.41%

Question 4

Using the same filters, which category has the lowest margin?

- Accessories
- Bikes
- Clothing
- Miscellaneous

Step 3:

Now, apply **Color scales** conditional formatting to **Margin** for the **Sub Category** level.

Question 5

Which Sub Category has the lowest margin?

- Caps
- Road Bikes
- Mountain Bikes
- Jerseys

Step 4:

Let's dig a little deeper. Add the **Product** field to the **Rows**, under **Sub Category**. Ensure that the **Year** and **Country** filter still select **2016** and **Australia** respectively. Apply **Color scales** conditional formatting only to the cells showing the **margin** for the **Product**.

Question 6

Which product has the least margin?

- AWC Logo Cap
- Road-350-W Yellow, 44
- Mountain-500 Silver, 42
- Short-Sleeve Classic Jersey, M

Module 6: Comparing Year over Year in Pivot Tables:

Before you start

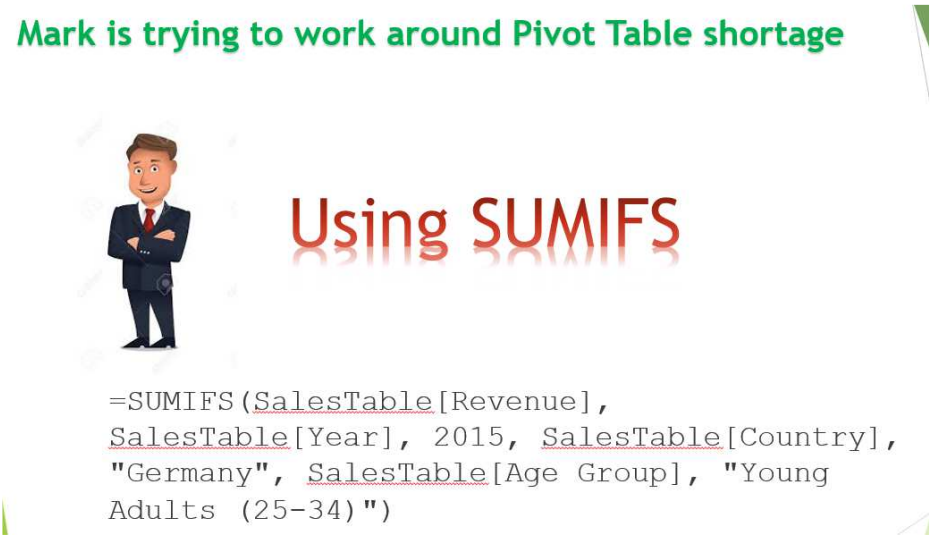
You need to know those formula in Excel :

- SUMIFS
- Spark line
- Icon Sets Conditional Formatting
  (Please Check Extra Demo Folder)

Demo 6 A: Sales Growth



Demo 6B: work around Pivot Table shortage



Knowledge Checks:

Question 1

What is the main advantage of referencing cells in a Pivot Table with the GetPivotData() function rather than by regular cell reference?

- The function returns a value that is always filtered by all slicers.
- It returns the same values even if the layout of the data changes.
- GetPivotData() is being refreshed automatically.
- GetPivotData() will display the values even if they are removed from the PivotTable

Question 2

What are the disadvantages of referencing cells in a Pivot Table with the GetPivotData() function and not by regular cell reference? Choose three that apply.

- The function is not adjusted when it is dragged across cells.
- It is very complicated to edit the function.
- It is very complicated to write the function manually.
- The function returns different values when the layout of the data changes

Question 3

What are the advantages of creating aggregates using SUMIFS() rather than using a Pivot Table? Choose three that apply.

- SUMIFS() formulas can be arranged in flexible formats in the grid.
- SUMIFS() formulas' comparison operators are not restricted to equality.
- SUMIFS() formulas are faster to calculate than creating aggregates.
- SUMIFS() formulas do not require refreshes to reflect changes in the data.

Question 4

What are the disadvantages of creating aggregates using SUMIFS() rather than using a Pivot Table? Choose three that apply.

- A range with SUMIFS formulas is much more difficult to change if you need to include more or fewer fields.
- SUMIFS() formulas cannot be based on an Excel table.
- Pivot Tables shrink and expand dynamically based on the data.
- On a report based on SUMIFS() formulas, you are unable to implement simple drill downs and drill ups.

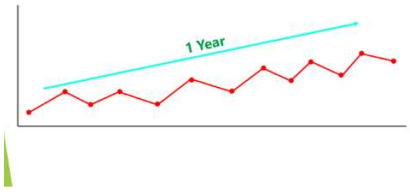
Lab 6:

Scenario:

Use Data in file Lab6Start.xlsx.

Lucy seems intrigued your presentation on Australia's sales.

Now, she wants to review the sales difference (growth) for Product Category and Sub Category between year 2015 and 2016. She wants to be able to filter the report by all the options available, including Customer Gender and Age Group.





You will create two reports. One with the sales for each year and the other with the % Change in Revenue from Year to Year. Each report is a cross-tabular format with **Product Categories** and **Sub categories** on the rows and **Year** on the columns, with **Sum of Revenue** as the aggregate data. When you are done, neither report will be based on a Pivot Table. However, you will enable a Pivot Table filter to allow the selection of **Country**, **Customer Gender**, and **Age Group** for these reports. Visually, you would like to have something like this:

Country	(All)
Customer Gender	(All)
Age Group	(All)

	2011	2012	2013	2014	2015	2016
Accessories						
Bike Racks						
Bike Stands						
Bottles and Cages						
Cleaners						
Fenders						
Helmets						
Hydration Packs						
Tires and Tubes						
Bikes						
Mountain Bikes						
Road Bikes						
Touring Bikes						
Clothing						
Caps						
Gloves						
Jerseys						
Shorts						
Socks						
Vests						

	2012	2013	2014	2015	2016
Accessories					
Bike Racks					
Bike Stands					
Bottles and Cages					
Cleaners					
Fenders					
Helmets					
Hydration Packs					
Tires and Tubes					
Bikes					
Mountain Bikes					
Road Bikes					
Touring Bikes					
Clothing					
Caps					
Gloves					
Jerseys					
Shorts					
Socks					
Vests					

Step 1:

Let's start by adding a new sheet named **Growth**. Starting in cell B1, add a new pivot table using SalesTable as the source with **Country**, **Customer Gender**, and **Age Group** as Filters, **Product Category** and **Sub Category** as Rows, **Year** as Columns, and **Sum of Revenue** as Values.

Step 2:

Now, copy the structure of the rows and columns to the cells next to the pivot table, that is, copy the **Years** labels in C6:H6 (excluding the Grand Total) to range L6:Q6, and the Product categories and Sub categories labels in B7:B26 (excluding the Grand Total) to range K7:K26. Highlight and copy range B6:H26, place your cursor in cell K6, right click and from the Paste Options, select Formatting (R).

Alternatively, you can select B6:H26, place your cursor in cell K6, right click and from the Paste Options, select Values (V) and then, once again right click in cell K6, and from the Paste Options, select Formatting (R). After which, delete the data area.

Step 3

Populate the data area of the newly created cross tabular structure using SUMIFS() function.
Assuming that you have:

- K7: Product Category eg "Accessories"
- Q6: Year eg. "2016"
- C1: Country Filter
- C2: Customer Gender Filter
- C3: Age Group Filter

Then, the cell that corresponds to **Year 2016** for the **Product Category Accessories** would have the following formula

```
=SUMIFS(SalesTable[Revenue],
SalesTable[Product Category],$K7,
SalesTable[Year],Q$6,
SalesTable[Country],IF($C$1="(All)","*",$C$1),
SalesTable[Customer Gender],IF($C$2="(All)","*",$C$2),
SalesTable[Age Group],IF($C$3="(All)","*",$C$3))
```

Step 4:

Now, copy this formula (using Ctrl C keyboard combination) and paste it (using Ctrl V keyboard combination) into the other Years and Product Category cells for all three categories (Accessories, Bikes, Clothing). When you are done, your calculated values in cells L7:Q7, L16:Q16 and L20:Q20 should match their respective pivot table values.

In addition, assuming that you have:

- K8: Sub Category eg "Bike Racks"
- Q6: Year eg. "2016"
- C1: Country Filter
- C2: Customer Gender Filter
- C3: Age Group Filter

Then, the cell that corresponds to **Year 2016** for the **Sub Category Bike Racks** would have the following formula

```
=SUMIFS(SalesTable[Revenue],
SalesTable[Sub Category],$K8,
SalesTable[Year],Q$6,
SalesTable[Country],IF($C$1="(All)","*",$C$1),
SalesTable[Customer Gender],IF($C$2="(All)","*",$C$2),
SalesTable[Age Group],IF($C$3="(All)","*",$C$3))
```

Now, copy this formula (using Ctrl C keyboard combination) and paste it (using Ctrl V keyboard combination) into the other Years and Sub Category cells. When you are done, your calculated values in these cells should match their respective pivot table values.

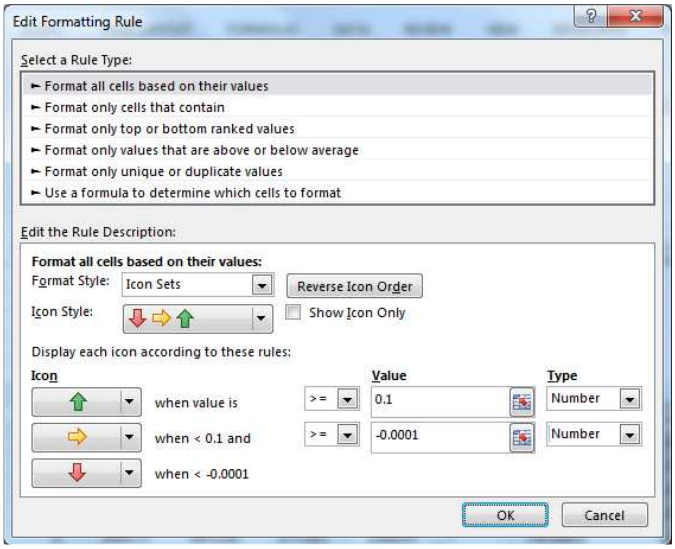
Once your table is populated, remove the fields from the Rows, Columns, and Values of the Pivot table, so that only the filters remains. Align the filters with your cross tabular report and add sparklines (to show the series high point) next to the column containing the last year of data.

When you are done with the first cross tabular report, create another one with the same structure next to it (excluding the 2011 column). This time, for % Change in Revenue, which is essentially the difference between the two years, divided by the previous year.
For example, one formula could be:

```
=IF(C7<>0,(D7-C7)/C7,0)
```

Step 5:

Present the data as percentage without decimal places. Apply an **Icon Sets** conditional formatting to differentiate positive and negative growth. Edit your conditional formatting icon set rules to display the icons similar to the rule below:



When you're done, you should have something like this (the below is just an example, your values may not necessarily match):

Country	(All)
Customer Gender	(All)
Age Group	(All)

	2011	2012	2013	2014	2015	2016
Accessories	0	0	4892974	6401902	4754492	6197992
Bike Racks	0	0	143388	207425	137420	195523
Bike Stands	0	0	122310	108545	113312	100045
Bottles and Cages	0	0	417227	614830	406739	597263
Cleaners	0	0	54387	75262	52785	73038
Fenders	0	0	359120	467332	350092	453042
Helmets	0	0	1641981	2204789	1597697	2133582
Hydration Packs	0	0	303315	344813	294578	336211
Tires and Tubes	0	0	1851246	2378906	1801869	2309288
Bikes	13832695	14024744	9106330	7352806	13660549	11042039
Mountain Bikes	3426187	3357010	3864069	3281283	5828199	4884435
Road Bikes	10406508	10667734	3860166	2224130	5780348	3328320
Touring Bikes	0	0	1382095	1847393	2052002	2829284
Clothing	0	0	2413273	3028443	2335355	2947394
Caps	0	0	145127	219016	140766	212625
Gloves	0	0	243801	342257	235500	332938
Jerseys	0	0	1196004	1529254	1159730	1489936
Shorts	0	0	544661	576724	527911	560849
Socks	0	0	38250	56560	37083	54863
Vests	0	0	245430	304632	234365	296183

	2012	2013	2014	2015	2016
Accessories	0%	0%	31%	-26%	30%
Bike Racks	0%	0%	45%	-34%	42%
Bike Stands	0%	0%	-11%	4%	-12%
Bottles and Cages	0%	0%	47%	-34%	47%
Cleaners	0%	0%	38%	-30%	38%
Fenders	0%	0%	30%	-25%	29%
Helmets	0%	0%	34%	-28%	34%
Hydration Packs	0%	0%	14%	-15%	14%
Tires and Tubes	0%	0%	29%	-24%	28%
Bikes	1%	-35%	-19%	86%	-19%
Mountain Bikes	-2%	15%	-15%	78%	-16%
Road Bikes	3%	-64%	-42%	160%	-42%
Touring Bikes	0%	0%	34%	11%	38%
Clothing	0%	0%	25%	-23%	26%
Caps	0%	0%	51%	-36%	51%
Gloves	0%	0%	40%	-31%	41%
Jerseys	0%	0%	28%	-24%	28%
Shorts	0%	0%	6%	-8%	6%
Socks	0%	0%	48%	-34%	48%
Vests	0%	0%	24%	-23%	26%

Question 1

Without applying any filter, which year does the **Accessories** category have negative growth?

- o 2013
- o 2014
- o 2015
- o 2016

Question 2

Filter the report for **Youth Age Group**. Which two years do the **Accessories** category have negative growth?

- o 2013
- o 2014
- o 2015
- o 2016

Question 3

Without applying any filter, which two years do the **Bikes** category have negative growth?

- o 2013
- o 2014
- o 2015
- o 2016

Question 4

Filter the report for **Australia**. Which year do the **Bikes** category have the highest growth?

- 2013
- 2014
- 2015
- 2016

Question 5

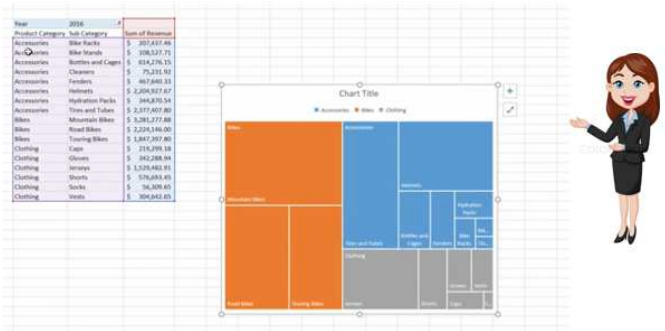
Keep the Australia filter. In the year that **Bikes** sales have the highest growth (previous question), which **Sub Category** of **Bikes** has the highest growth?

- Mountain Bikes
- Road Bikes
- Touring Bikes
- City Bikes

Module 7: Reporting Hierarchical Data

Demo 1: Charting Capabilities:

Treemap



Sunburst



Knowledge Checks

Question 1

How would you modify the following formula to include age ranges in the following report?

```
=SUMIFS(SalesTable[Revenue],SalesTable[Customer
Gender],B$4,SalesTable[Country],$A5)
```

	A	B	C
1	From Age:	19	
2	To Age:	38	
3			
4	Country \ Gender	F	M
5	France	1,675,738	1,770,366
6	Germany	2,181,725	2,062,785
7	United Kingdom	2,076,685	2,199,535
8	United States	4,998,486	5,379,256

- `=SUMIFS(SalesTable[Revenue],SalesTable[Customer Gender],B$4,SalesTable[Country],$A5,SalesTable[Customer Age],">=$B$1",SalesTable[Customer Age],"<=$B$2")`

- `=SUMIFS(SalesTable[Revenue],SalesTable[Customer Gender],B$4,SalesTable[Country],$A5,SalesTable[Customer Age], AND(SalesTable[Customer Age] >=$B$1,SalesTable[Customer Age]<=$B$2 ))`

- `=SUMIFS(SalesTable[Revenue],SalesTable[Customer Gender],B$4,SalesTable[Country],$A5,SalesTable[Customer Age],">="&$B$1,SalesTable[Customer Age],"<="&$B$2)`

- `=SUMIFS(SalesTable[Revenue],SalesTable[Customer Gender],B$4,SalesTable[Country],$A5,SalesTable[Customer Age],">="&$B$1 AND SalesTable[Customer Age],"<="&$B$2)`

Question 2

What is the main purpose of using the Treemap and Sunburst charts to display data? Select the best answer.

- To compare values between elements.
- To use colors to represent values.
- To show the relative size at different levels of a hierarchy.
- To show trend in data.

Question 3

You want to create a SUMIFS() formula that references a specific cell in your spreadsheet as you apply it to a range of data containing multiple columns. What should you do?

- Use Name Manager, and use the names in your formula.
- Use referenced table columns by name.
- Write a separate formula for each column, and then copy/paste the formula to all rows.
- Use a mix of absolute and relative reference in your formula.

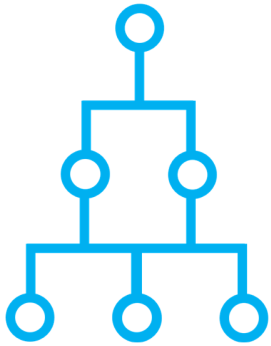
Lab 7: Reporting hierarchal data

Use Data in file Lab7Start.xlsx.

Scenario

Upon reviewing the growth report you created, Lucy asked for a report that shows composition of Product Categories and Sub Categories based on certain filters, including Year, Country, Customer Gender, and Age Group.

Specifically, Lucy wants to see the report visualized using a hierarchical chart.



Step 1:

Let's add another sheet to the workbook and name it **Composition**.

Start by adding a new pivot table with **Year, Country, Customer Gender**, and **Age Group** as Filters, **Product Category** and **Sub Category** as Rows, and **Sum of Revenue** as Values.

Format the pivot table to show in **Tabular** form, **do not show subtotals** and **grand totals**, and **repeat all item labels**.

Step 2:

Now, copy the structure of the rows and columns to the cells next to the pivot table, that is the **Product categories** and **Sub categories** as Rows. Delete the data area for now.

Populate the data area of the newly created cross tabular structure using SUMIFS() function.
Assuming that you have:

- F7: Product Category eg. "Accessories"
- G7: Sub Category eg. "Bike Racks"
- C1: Year filter
- C2: Country Filter
- C3: Customer Gender Filter
- C4: Age Group Filter

Then, the cell that corresponds the **Product Category Accessories** and **Sub Category Bike Racks** would have the following formula:

```
=SUMIFS(SalesTable[Revenue],
SalesTable[Product Category],$F7,
SalesTable[Sub Category],$G7,
SalesTable[Year],IF($C$1="(All)",">0",$C$1),
SalesTable[Country],IF($C$2="(All)","*",$C$2),
SalesTable[Customer Gender],IF($C$3="(All)","*",$C$3),
SalesTable[Age Group],IF($C$4="(All)","*",$C$4))
```

Step 3:

Once your table is populated, remove the fields from the Rows, Columns, and Values of the Pivot table, so that only the filters remains. Align the filters with your cross tabular report. Create a treemap or sunburst chart based on the data in the cross tabular report. Name the chart "Sales Composition".

Question 1

Explore the sales composition of **Bikes** category for each **Age Group**.
Which **Age Group** does the composition (rank of sales) differ than the rest?

- Youth
- Young Adult
- Adults
- Seniors
- The composition are the same across Age Group

Question 2

0/1 point (graded)

Now explore the sales composition of **Bikes** category for each **Age Group**, for the **Male** customers. Which **Age Group** does the composition differ than the rest?

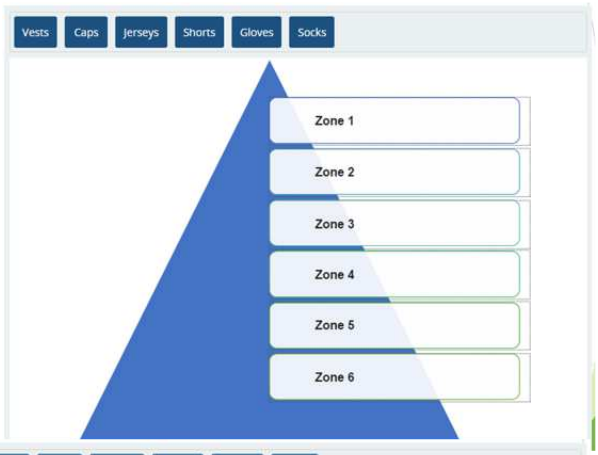
- Youth
- Young Adult
- Adults
- Seniors
- The composition are the same across Age Group

Question 3

PROBLEM

Clear all filters. Now, filter for the year **2016** and **Germany**. Rank the sales from the highest to lowest for the **Clothing** category.

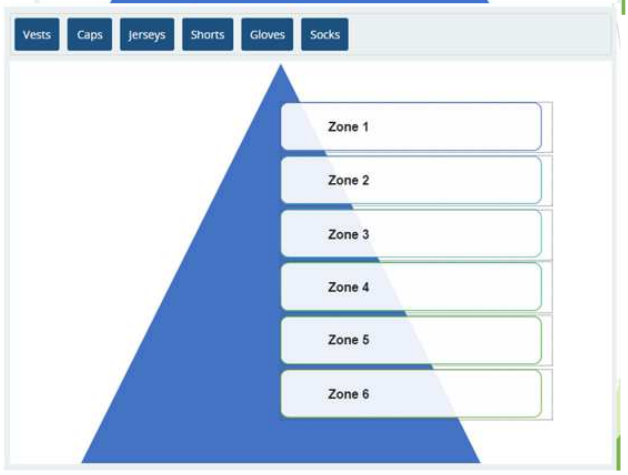
HINT: You might want to specify the data for the charts only for the **Clothing** category.



Question 4

PROBLEM

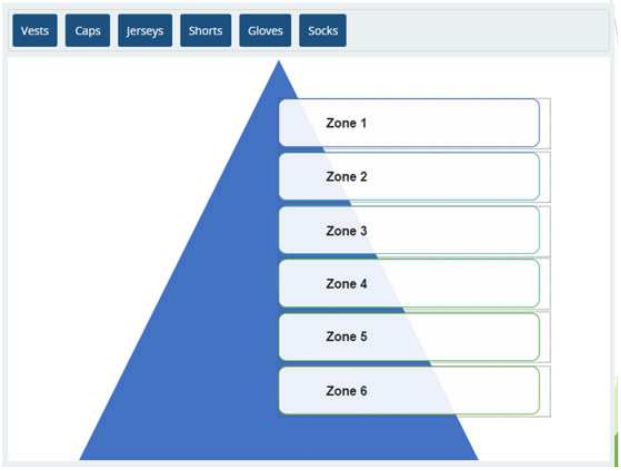
Keep the filter settings and add filter by **Male** customers. Rank the sales from the highest to lowest for the **Clothing** category



Question 5

PROBLEM

Keep the filter settings and add filter by **Youth Age Group**. Rank the sales from the highest to lowest for the **Clothing** category.



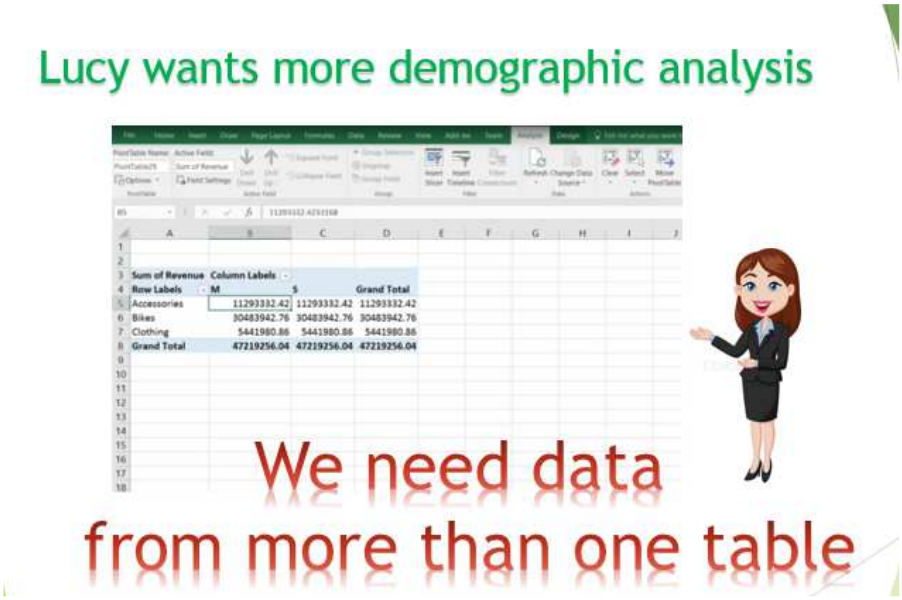
Module 8: Introduction to multi-table pivot tables and the data model

Before you start

You must Know this formula in Excel :

- VLOOKUP

Demo 8: Data from more than one table



Knowledge Checks

Question 1

You want combine data in two ranges/tables in a single pivot. What should you do? Choose two that apply.

- Click ‘More Tables…’ from the pivot field list, and create relationships between the different ranges.
- Connect the two sources of data using the same slicer.

- Copy the second range, append to the first, and define the pivot on the combined range.
- Combine the different ranges into one range using VLOOKUP formulas.

Lab 8: Data from multiple tables

Use Data in file_Lab8Start.xlsx. And……..Demographics.mdb or (Customer_demographics.txt)

Scenario

Lastly, Lucy wants to have more information about the customer demographics in addition to the already available age and gender. Since your data has Customer ID for each row, you can "connect" these rows with your customer demographics database. Your customer demographics is stored in an Access database.

Step 1:

Let's bring the customer database into your Excel.

1. From a new worksheet, go to the **Data** tab and on the **Get External Data** group, select the **From Access** option. This will bring up the **Select Data Source** dialog box.

2. Navigate to where you saved the **Demographics.mdb** file, select it and click on the **Open** button.

3. From the **Import Data** dialog box, ensure that **Table** is selected for how you want to view the data, within the existing worksheet (or select New Worksheet if you didn't start from a new one), then click on the **OK** button.

4. There is only one table in this database so the import is rather simple. Rename the worksheet **Demographic Data**. Your import should result in 18,484 rows (plus the header) and seven columns of data.

5. Create a new pivot table **from the SalesTable** then click **More tables**. Remember to select **Yes** to Excel's prompt to create a new pivot table.

6. Now, add the **MaritalStatus** and **TotalChildren** fields from the **Customer_demographic** table as **Rows** and **Order Quantity** from the **SalesTable** as **Values**. In addition, add the **Product Category** as filters.

Question 1

For those customers who bought bikes, what are the top **three** (bought the most quantity) customer profiles (marital status and number of children)?

- Single without child
- Single with one child
- Married with one child
- Married with two children

Step 2:

Now, remove the **MaritalStatus** and **TotalChildren** fields from the Rows and replace them with the **YearlyIncome** field from the **Customer_demographic** table.

Question 2

PROBLEM

For those customers who bought bikes, what are the top three (bought the most quantity) income brackets? (Rank from highest to Lowest)

HINT: Sort the **YearlyIncome** by **Sum of Order Quantity** in Descending order.



Step 3:

Now, remove the YearlyIncome field from the Rows and replace it with the **EnglishEducation** field from the **Customer_demographic** table.

Question 3

For those customers who bought bikes, what are the top **two** (bought the most quantity) education levels?

- High School
- Partial College
- Bachelors
- Graduate Degree

Step 4:

Lastly, replace the Rows with the **HouseOwnerFlag** field from the **Customer_demographics** table. Format the Sum of Order Quantity to show as Percentage of Grand Total with two decimal places.

Lastly, remove the **EnglishEducation** field from the Rows and replace it with the **HouseOwnerFlag** field from the **Customer_demographics** table. Format the **Sum of Order Quantity** to show as Percentage of Grand Total with two decimal places.

Question 4

What is the percentage of the customers who bought **Bikes** and are house owners?

- 31.28%
- 38.72%
- 61.28%
- 68.72%